

Subsurface Convection as the Origin of Stochastic Photometric Variability in Massive Stars

MATTEO CANTIELLO,^{1,2} DANIEL LECOANET,² AND ADAM S. JERMYN¹

¹*Center for Computational Astrophysics, Flatiron Institute, New York, NY 10010, USA*

²*Department of Astrophysical Sciences, Princeton University, Princeton, NJ 08544, USA*

Submitted to ApJ

ABSTRACT

Recent high-precision photometric observations have revealed a new ubiquitous phenomena in early type stars: stochastic low-frequency photometric variability. There are currently two main ideas for the origin of this photometric variability: Internal Gravity Waves (IGW) launched by the convective core, or granulation/IGW produced by (sub)surface convection. Thanks to high resolution ground-based spectroscopy, many of the targets observed with K2 and TESS now have also good stellar parameters, including mass, luminosity and a measure of surface velocity fluctuations (macroturbulence). Here we show theoretical predictions for the trend of characteristic frequency and amplitude of the perturbation induced by subsurface convection in massive stars, and compare these predictions to observations. We found that both the characteristic frequency ν_{char} and amplitude α_0 of the observed photometric variability correlate very well with the typical frequency τ_c^{-1} and flux of IGWs launched by the iron subsurface convection zone (FeCZ), respectively. While the amplitude of photometric variability also correlates with theoretical predictions of the flux of IGWs generated by core convection, the observed trend of characteristic frequency is at odds with theoretical expectations from convective turnover timescales in convective cores. This tentatively breaks the degeneracy between the two plausible models in favor of subsurface convection. We suggest a unified picture in which low-frequency photometric variability, surface turbulence, magnetic spots and wind variability in massive stars could all have the same origin: the presence of subsurface convection.

1. INTRODUCTION

Space Photometry revolution. New type of variability. Other mysteries at the surface of massive stars (turbulence, wind variability, spots). Stellar structure: convection in the core and convection in the envelope. IGWs and granulation. We discuss the properties of subsurface convection, in particular the maximum amplitude of waves that can be excited by their turbulent activity and their frequency as a function of the stellar temperature and luminosity. We show that the predicted trends of IGWs excited by turbulent convection in the FeCZ match very well the observations. We then proceed showing that while the trend with amplitude of the photometric variability could be reproduced by IGWs launched by core convection, the trend with excitation frequency in the core is opposite to the one observed at the surface. We discuss if wave propagation through

the stellar envelope can in principle revert this trend, and conclude this is unlikely. We conclude suggesting that subsurface convection represents a possible unifying mechanism for low-frequency photometric variability, surface turbulence, magnetic spots and wind variability in massive stars

This shows up as stochastic low-frequency variability that was observed in a large fraction of massive OB stars (see e.g. Balona 1992; Buysschaert et al. 2015; Bowman et al. 2019b).

2. METHODS

We calculated stellar evolution models using the Modules for Experiments in Stellar Astrophysics (MESA Paxton et al. 2011, 2013, 2015, 2018, 2019) software instrument. Details on the microphysics inputs to this software instrument are given in Appendix A. The inlists used to run our models can be found in this [GitHub repository](#)¹. We focused on the structure and con-

Corresponding author: Matteo Cantiello
mcantiello@flatironinstitute.org

¹ <https://github.com/matteocantiello/rednoise>

vective properties of our models, in order to calculate typical frequencies and convective fluxes. Convection is calculated in the framework of the Mixing Length Theory (Böhm-Vitense 1958, MLT hereafter), and we adopted $\alpha_{\text{MLT}} = 1.6$. While in efficient convection zones (e.g. stellar cores) the choice of this parameter does not change results much, this is not true in superadiabatic, inefficient convective regions close to the stellar surface (Jiang et al. 2015; Cantiello & Braithwaite 2019). We discuss in Section 3.4 how uncertainties in the choice of the α_{MLT} parameter affect our results.

Since we are interested in the excitation of surface phenomena, we focus on the upper part of convective zones and following Cantiello et al. (2009) we define the average of a quantity $\bar{q}$ as

$$\bar{q} \equiv \frac{1}{\alpha_{\text{MLT}} H_{\text{P}}} \int_{R_{\text{c}} - \alpha_{\text{MLT}} H_{\text{P}}}^{R_{\text{c}}} q(r) dr \quad (1)$$

where H_{P} is the pressure scale height calculated at the upper boundary (R_{c}) of the convective zone of interest. We tested using different average procedures, and found that our results do not depend much on the specific choice.

Using this definition, we extracted the average Mach number $\bar{M}_{\text{c}}$, convective velocity $\bar{v}_{\text{c}}$ and density $\bar{\rho}_{\text{c}}$ in the convective core and in the subsurface convection zones of our models. The most important subsurface convection zone for the massive stars discussed in this work is the Iron Convection Zone (FeCZ) (Cantiello et al. 2009), although in the low luminosity regime He-driven convection zones could play a role as well (Cantiello & Braithwaite 2019). We define the characteristic frequency of convection as

$$\nu_{\text{char}} \equiv \frac{1}{2\pi\tau_{\text{c}}}, \quad (2)$$

where $\tau_{\text{c}} = \alpha_{\text{MLT}} \bar{H}_{\text{P}} / \bar{v}_{\text{c}}$ is the average convective turnover time in a convection zone. We proceeded to calculate convective turnover timescales and other relevant properties like convective fluxes $F_{\text{c}} = \bar{\rho}_{\text{c}} \bar{v}_{\text{c}}^3$. We did this as a function of both mass and evolutionary history for stars ranging from $5 - 120 M_{\odot}$. Here we present results for an initial metallicity of $Z=0.02$, but in Appendix B we report results for model grids with $Z = 0.014, 0.006$ and 0.002 as well.

We compare these quantities with the observed characteristic frequency ν_{char} and amplitude α_0 of stochastic photometric variability. These quantities are derived by fitting the stochastically-excited low-frequency power excess in a power density spectrum using a Lorentzian function (e.g. Bowman et al. 2019a):

$$\alpha(\nu) = \frac{\alpha_0}{1 + \left(\frac{\nu}{\nu_{\text{char}}}\right)^{\gamma}} + P_{\text{W}}. \quad (3)$$

This shows that α_0 represents the amplitude at zero frequency, and ν_{char} is the frequency at which the amplitude of background equals half of α_0 . Finally, γ is the gradient of the linear part of the profile and P_{W} is a white noise term term.

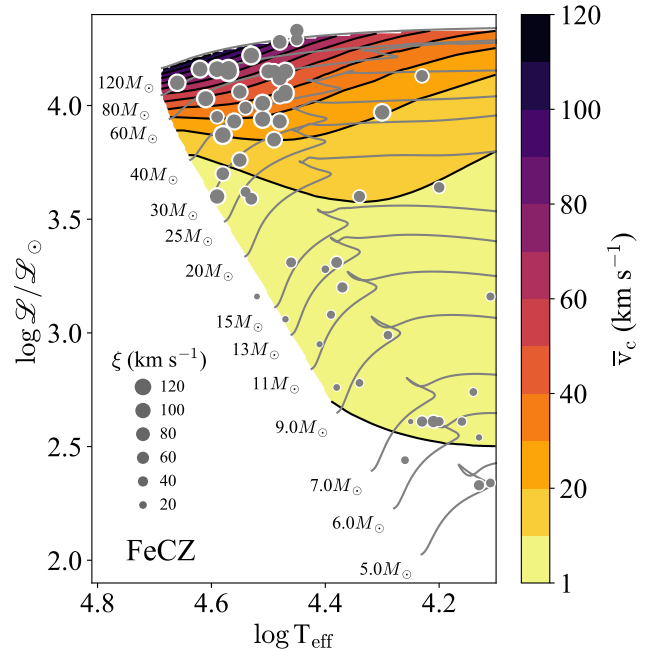


Figure 1. Average convective velocities in the FeCZ as function of the location of stellar models in the spectroscopic H-R Diagram. We also show the observed stars with detected macroturbulent velocity as grey dots. The size of the dot is proportional to the observed macroturbulence.

3. RESULTS

We want to test a possible correlation between the properties of the FeCZ and observed photospheric phenomena, in particular photometric variability and turbulent velocities at the stellar surface (macroturbulence). If photometric variability was caused by the FeCZ, a natural choice of proxy for the typical timescale of this variability is the convective turnover timescale. We found typical values of the convective turnover timescale to be about $\sim 0.1 \dots 2$ d in the FeCZ, with a tendency for shorter values in models with high effective temperature and surface luminosity (Cantiello et al. 2009). To quantitatively compare to the characteristic frequencies of the observed stochastic photometric variability in massive stars, we proceed mapping our theoretical predictions for the quantity ν_{char} (eq. 2) on a spectroscopic H-R diagram (Langer & Kudritzki 2014). This reveals a very good agreement, with our models reproducing both the typical values of the observed ν_{char}

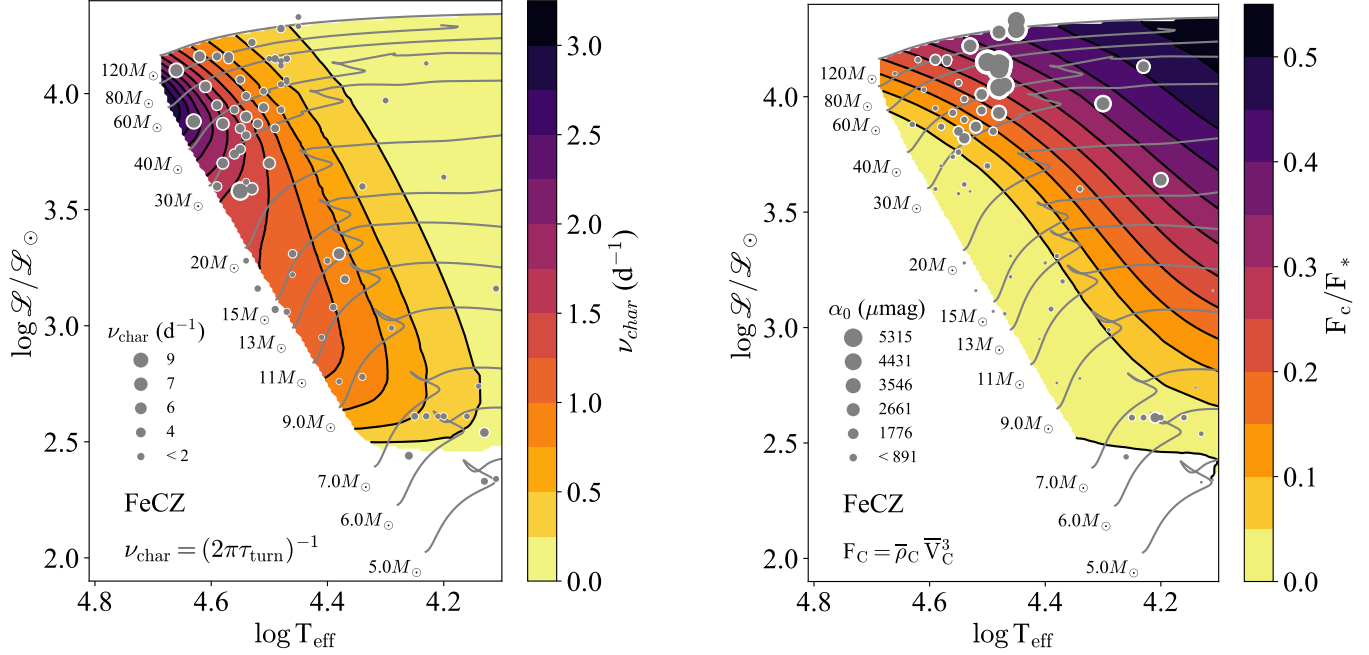


Figure 2. Left panel: Characteristic frequency ν_{char} in the FeCZ as a function of the location of stellar models in the spectroscopic H-R Diagram. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} . Right panel: Luminosity fluctuations expected due to IGWs launched by the FeCZ. This scales with the square root of the relative IGWs flux, with a correction factor accounting for the likely scale of the perturbation. We assumed uncorrelated perturbations at the surface with typical scale of a convective cell, so a correction factor $\sim 1/\sqrt{N_{\text{Cells}}}$. The IGWs flux is calculated multiplying the convective flux in the FeCZ by the average convective Mach number (this is calculated in the top pressure scale height of the convective region. The agreement is pretty good, considering that uncertainties of a factor of 2 in the MLT α parameters for these regions result in a factor of 20 in luminosity perturbations). We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

and the trend with $\log \mathcal{L}$ and $\log T_{\text{eff}}$ (Fig. 2). While the characteristic frequencies found by Bowman20 seem to be larger by a factor of ≈ 3 , our predictions for the turnover timescale are affected by uncertainty in the convective velocities arising from the MLT treatment ($v_c \propto \alpha_{\text{MLT}}^3$, so $\nu_{\text{char}} \propto \alpha_{\text{MLT}}^2$). Since α_{MLT} is uncertain by a factor of 2 or so, our estimates of the characteristic frequency in the FeCZ are uncertain by a factor of ≈ 4 .

We proceed with mapping quantities that measure the amplitude of perturbations at the location of the FeCZ. We then check if some of these properties correlate with the amplitude of observed photospheric phenomena, including turbulent velocity fluctuations and photometric variability. While here we are agnostic about the particular mechanism, we assume that some physical process connects the subsurface layers to the photosphere. We discuss the most likely candidates for this process in Sec. 3.5. We show the average convective velocity in the FeCZ in Fig. 2. Velocities of the order $10 \dots 100 \text{ km s}^{-1}$ are found across the upper H-R diagram, with a trend of increasing v_c for higher luminosities. This trend is similar to what is observed for the macroturbulent velocity

field at the surface of massive stars. This is in agreement with the results of Grassitelli et al. (2015), who found a tight correlation between the turbulent pressure ($P_{\text{turb}} \propto v_c^2$) in the FeCZ and spectroscopically-derived macroturbulent velocities in massive stars. Interestingly, the range of the velocity field in the FeCZ is also similar to the observed range of macroturbulent velocities. Again the caveat is the large uncertainty (a factor ≈ 10) of the convective velocities calculated using the MLT in inefficient convective zones (Cantiello & Braithwaite 2019). We plot the average Mach number in the FeCZ in Fig. 3.2, showing that contrary to core convection the FeCZ is just mildly subsonic, with values of the Mach number ranging from 0.01 to 0.3. The relative convective flux in these regions is shown in Fig. 2, where we also report the the observed amplitude of photometric variability. Values of F_c/F_* tend to increase with increasing $\log \mathcal{L}$ and decreasing $\log T_{\text{eff}}$, a trend that is also found for the values of observed photometric variability.

3.1. Core Convection and Photometric Variability

Early-type stars possess convective cores during their main sequence. Turbulent convection in these convective cores excites internal gravity waves (Goldreich & Kumar 1990; Rogers et al. 2013; Lecoanet & Quataert 2013; Shiode et al. 2013; Edelmann et al. 2019), that can propagate through the stellar envelope and reach the stellar surface. These waves could in principle be responsible for both the observed macroturbulence and photometric variability in early-type stars (Aerts et al. 2009; Simón-Díaz et al. 2017; Bowman et al. 2019a,b). However, Lecoanet et al. (2019) studied the effects of wave propagation and radiative damping in the stellar envelope, concluding that it is unlikely that the observed photometric variability is caused by internal gravity waves launched by the convective core. One of the biggest uncertainties is the excitation spectrum of internal gravity waves. While Coustou et al. (2018) found... Edelmann et al. (2019) claims that...

Here we focus on the core convective properties of our models, and similarly to the case of the FeCZ we look for correlations with the amplitude and characteristic frequency of the observed surface variability. The flux of Internal Gravity Waves excited by turbulent convection at the core boundary is $F_{\text{IGW}} = F_* \bar{M}_c$ (Goldreich & Kumar 1990). We expect the surface radial displacement ξ_r/R produced by internal gravity waves launched from the core to be proportional to the quantity $\sqrt{F_{\text{IGW}}/F_*}$. We show this quantity in Fig. 7. The Mach number in the convective core is on the order of $10^{-3} \dots 10^{-4}$, with a trend of increasing values for lower $\log T_{\text{eff}}$ and higher $\log \mathcal{L}$. This trend matches the observed trend in the photometric variability pretty well.

Next, we focus on the characteristic timescale of waves excited by core convection. We expect the maximum of the IGW flux to be launched at frequencies close to $\nu_{\text{char}} = (2\pi\tau_c)^{-1}$. Fig. 7 shows that the predicted values of ν_{char} are in the range $0.02 \dots 0.008 \text{ d}^{-1}$ ($0.2 \dots 0.08 \mu\text{Hz}$), in agreement with results of Shiode et al. (2013). These values are about 2 orders of magnitude smaller than the typical characteristic frequencies of stochastic photometric variability observed by (Bowman et al. 2019a,b). Wave propagation through the stellar envelope affects the spectrum, changing the frequency of maximum power of waves reaching the surface (Lecoanet et al. 2019). Radiative damping in particular can efficiently suppress waves at low frequency, so that the peak of the spectrum observed at the surface is expected to move to higher frequencies. The amplitude of this effect depends on the envelope properties, which are a function of $\log T_{\text{eff}}$ and $\log \mathcal{L}$. However, we expect the trends of predicted ν_{char} as function of $\log T_{\text{eff}}$ and $\log \mathcal{L}$ to be mostly determined by the core convective

properties (Daniel check / expand on this?). In Fig. 7 the prediction is that ν_{char} should increase with both $\log T_{\text{eff}}$ and $\log \mathcal{L}$, so that the characteristic frequencies of peak IGWs should increase as stars evolve on the main sequence. This is exactly the opposite of what is observed: the characteristic frequencies of stochastic photometric variability is largest for stars early on on the main sequence, and seem to decrease as stars evolve. Therefore the data seems to break the degeneracy between the proposed models in favor of subsurface convection.

3.2. Macroturbulence

Discuss here magnetic inhibition as a way to distinguish subsurface vs core generation (Adam's calculations). The hypothesis that subsurface convection is responsible for surface turbulence is corroborated by the match between trends in the FeCZ properties (convective velocities / turbulent pressure) and the observed macroturbulence (Cantiello et al. 2009; Grassitelli et al. 2015). A further test of this idea was provided by ?, who showed that only stars with a magnetic fields strong enough to suppress the FeCZ show surface velocity fields consistent with no macroturbulence. Of course one could envision that magnetic inhibition could also inhibit the propagation of IGWs coming from the convective core. Below we show that the minimum magnetic field required to stop core IGWs from reaching the stellar surface is much smaller than what required to shutoff the FeCZ. This provides a way to test the two hypothesis...

3.3. A unified model for surface phenomena in massive stars

The presence of subsurface convection can simultaneously account for a large variety of puzzling phenomena that appear ubiquitous at the surface of early-type stars. First of all we have microturbulence and macroturbulence. Both can be accounted for by velocity fields excited by the underlying FeCZ subsurface convection zone (Cantiello 2009, Grassitelli 2015, This work). Photospheric magnetic spots: These spots might cause the ubiquitous DACs, resulting from CIRs in the winds of massive stars. The seed of wind clumping. Finally, the presence of a turbulent velocity field necessarily has to induce photometric variability, and we have shown here that the observed low-frequency stochastic noise observed in the majority of massive stars correlates very well with the presence and properties of the FeCZ. As such, a minimal hypothesis is that the presence of the FeCZ is the underlying physical cause and represent a possible unified model for the appearance of

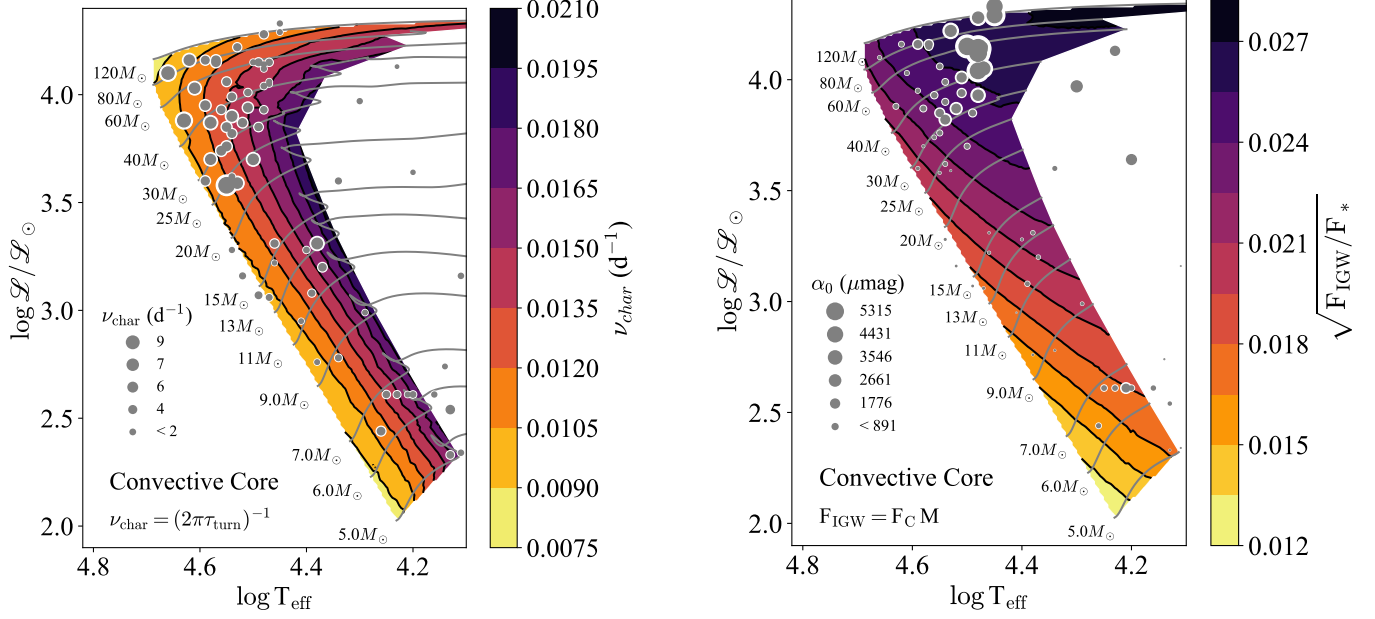


Figure 3. Left panel: Characteristic frequency ν_{char} in the convective core as function of the location of stellar models in the spectroscopic HRD. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} . Right panel: Flux of gravity waves launched by the convective core. This is calculated multiplying the convective flux in the core by the average convective Mach number (this is calculated in the top pressure scale height of the convective region). We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

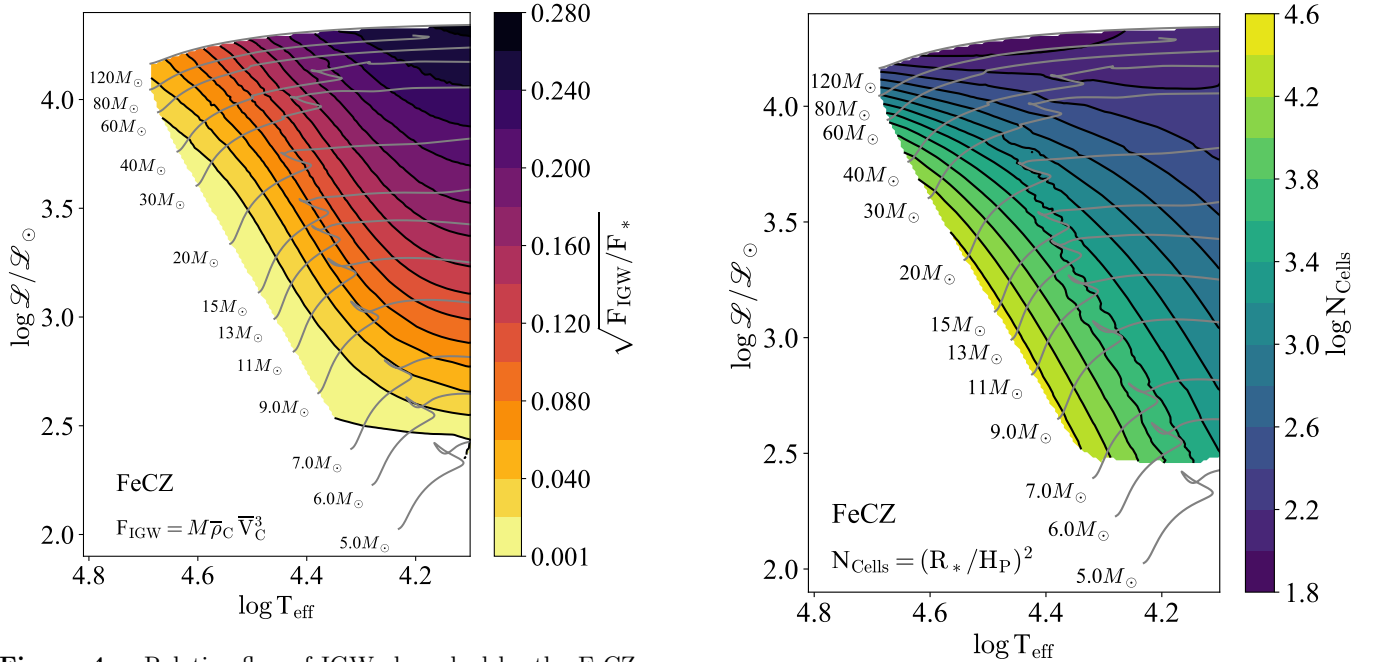


Figure 4. Relative flux of IGWs launched by the FeCZ. The IGWs flux is calculated multiplying the convective flux in the FeCZ by the average convective Mach number, calculated in the top pressure scale height of the convective region.

Figure 5. Expected number of convective cells in the FeCZ. This is calculated assuming the size of convective cells is about the average pressure scale height in the FeCZ

turbulence, magnetic spots, low-frequency photometric

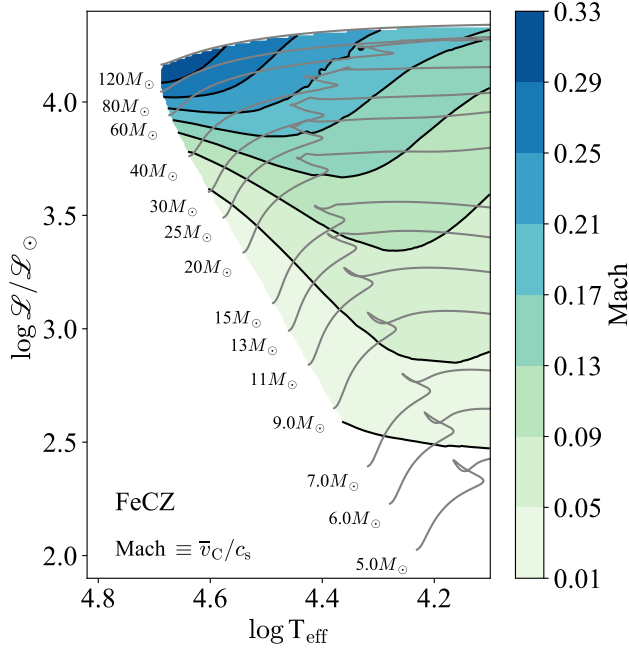


Figure 6. Mach number in the FeCZ.

variability and wind structure and variability in massive stars.

3.4. Discussion

3.5. Above the Iron Convection Zone

The presence of a subsurface convection zone can result in variability of photospheric properties. The simplified MLT treatment in our one dimensional calculations is unlikely to capture the complex phenomenology of these layers. Multi dimensional simulation including radiation have been performed in a restricted range of the parameter space, and show that the full turbulent manifestation of these convective regions extends up to the stellar surface (Jiang et al. 2015, 2018). This can occur both because turbulent velocities excited by the convection extends to

3.6. The Impact of α_{MLT}

3.7. Conclusions

ACKNOWLEDGMENTS

The Center for Computational Astrophysics at the Flatiron Institute is supported by the Simons Foundation. This research was supported in part by the National Science Foundation under Grant No. NSF PHY-1748958 and by the Gordon and Betty Moore Foundation through Grant GBMF7392

APPENDIX

A. SOFTWARE DETAILS

Calculations were done with MESA version 11701. The MESA EOS is a blend of the OPAL Rogers & Nayfonov (2002), SCVH Saumon et al. (1995), PTEH Pols et al. (1995), HELM Timmes & Swesty (2000), and PC Potekhin & Chabrier (2010) EOSes.

Radiative opacities are primarily from OPAL (Iglesias & Rogers 1993, 1996), with low-temperature data from Ferguson et al. (2005) and the high-temperature, Compton-scattering dominated regime by Buchler & Yueh (1976). Electron conduction opacities are from Cassisi et al. (2007).

Nuclear reaction rates are a combination of rates from NACRE (Angulo et al. 1999), JINA REACLIB (Cyburt et al. 2010), plus additional tabulated weak reaction rates Fuller et al. (1985); Oda et al. (1994); Langanke & Martínez-Pinedo (2000). Screening is included via the prescription of Chugunov et al. (2007). Thermal neutrino loss rates are from Itoh et al. (1996).

The inlists, processing scripts, and model output are available at [Zenodo.org](https://zenodo.org).

B. GRIDS AT DIFFERENT METALLICITIES

Here we present results for the critical and surface magnetic field at metallicities of $Z=0.014$, $Z=0.006$, and $Z=0.002$, representing early-type stars in the Galaxy (MW), Large Magellanic Cloud (LMC), and Small Magellanic Cloud (SMC) respectively (e.g. Yusof et al. 2013). The initial Helium content of the grids is $Y=0.2659$ (MW), $Y=0.2559$ (LMC), and $Y=0.2508$ (SMC). The metallicity is initialized scaling the standard solar composition of Grevesse & Sauval (1998).

REFERENCES

- Aerts, C., Puls, J., Godart, M., & Dupret, M. A. 2009, A&A, 508, 409, doi: [10.1051/0004-6361/200810471](https://doi.org/10.1051/0004-6361/200810471)
- Angulo, C., Arnould, M., Rayet, M., et al. 1999, Nuclear Physics A, 656, 3, doi: [10.1016/S0375-9474\(99\)00030-5](https://doi.org/10.1016/S0375-9474(99)00030-5)

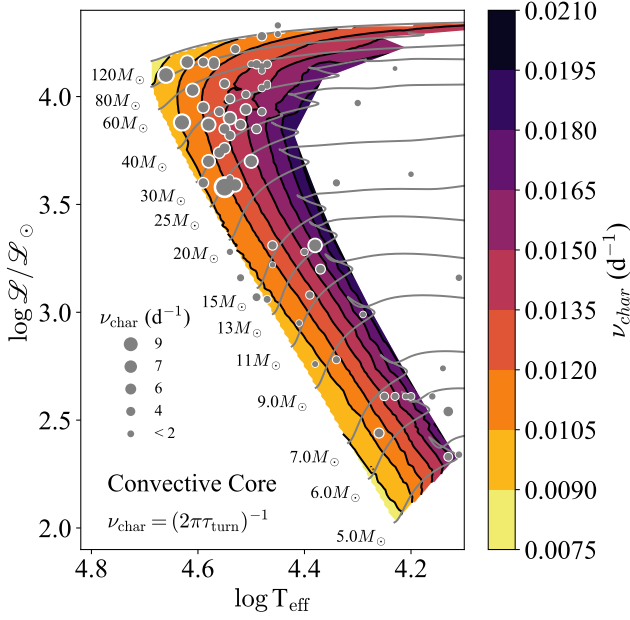


Figure 7. Characteristic frequency ν_{char} in the convective core as function of the location of stellar models in the spectroscopic HRD. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} .

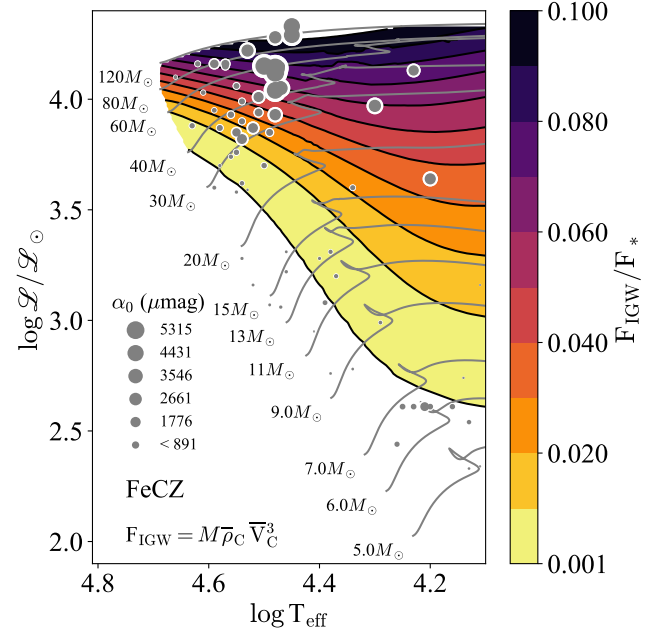


Figure 8. Flux of gravity waves launched by the FeCZ. This is calculated multiplying the convective flux in the FeCZ by the average convective Mach number (this is calculated in the top pressure scale height of the convective region). We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

Böhm-Vitense, E. 1958, ZA, 46, 108

Bowman, D. M., Aerts, C., Johnston, C., et al. 2019a, A&A, 621, A135, doi: [10.1051/0004-6361/201833662](https://doi.org/10.1051/0004-6361/201833662)

Bowman, D. M., Burssens, S., Pedersen, M. G., et al. 2019b, Nature Astronomy, 3, 760, doi: [10.1038/s41550-019-0768-1](https://doi.org/10.1038/s41550-019-0768-1)

Buchler, J. R., & Yueh, W. R. 1976, ApJ, 210, 440, doi: [10.1086/154847](https://doi.org/10.1086/154847)

Cantiello, M., & Braithwaite, J. 2019, ApJ, 883, 106, doi: [10.3847/1538-4357/ab3924](https://doi.org/10.3847/1538-4357/ab3924)

Cantiello, M., Langer, N., Brott, I., et al. 2009, A&A, 499, 279, doi: [10.1051/0004-6361/200911643](https://doi.org/10.1051/0004-6361/200911643)

Cassisi, S., Potekhin, A. Y., Pietrinferni, A., Catelan, M., & Salaris, M. 2007, ApJ, 661, 1094, doi: [10.1086/516819](https://doi.org/10.1086/516819)

Chugunov, A. I., Dewitt, H. E., & Yakovlev, D. G. 2007, PhRvD, 76, 025028, doi: [10.1103/PhysRevD.76.025028](https://doi.org/10.1103/PhysRevD.76.025028)

Couston, L.-A., Lecoanet, D., Favier, B., & Le Bars, M. 2018, Journal of Fluid Mechanics, 854, R3, doi: [10.1017/jfm.2018.669](https://doi.org/10.1017/jfm.2018.669)

Cyburt, R. H., Amthor, A. M., Ferguson, R., et al. 2010, ApJS, 189, 240, doi: [10.1088/0067-0049/189/1/240](https://doi.org/10.1088/0067-0049/189/1/240)

Edelmann, P. V. F., Ratnasingam, R. P., Pedersen, M. G., et al. 2019, ApJ, 876, 4, doi: [10.3847/1538-4357/ab12df](https://doi.org/10.3847/1538-4357/ab12df)

Ferguson, J. W., Alexander, D. R., Allard, F., et al. 2005, ApJ, 623, 585, doi: [10.1086/428642](https://doi.org/10.1086/428642)

Fuller, G. M., Fowler, W. A., & Newman, M. J. 1985, ApJ, 293, 1, doi: [10.1086/163208](https://doi.org/10.1086/163208)

Goldreich, P., & Kumar, P. 1990, ApJ, 363, 694, doi: [10.1086/169376](https://doi.org/10.1086/169376)

Grassitelli, L., Fossati, L., Simón-Díaz, S., et al. 2015, ApJL, 808, L31, doi: [10.1088/2041-8205/808/1/L31](https://doi.org/10.1088/2041-8205/808/1/L31)

Grevesse, N., & Sauval, A. J. 1998, SSRv, 85, 161, doi: [10.1023/A:1005161325181](https://doi.org/10.1023/A:1005161325181)

Iglesias, C. A., & Rogers, F. J. 1993, ApJ, 412, 752, doi: [10.1086/172958](https://doi.org/10.1086/172958)

—. 1996, ApJ, 464, 943, doi: [10.1086/177381](https://doi.org/10.1086/177381)

Itoh, N., Hayashi, H., Nishikawa, A., & Kohyama, Y. 1996, ApJS, 102, 411, doi: [10.1086/192264](https://doi.org/10.1086/192264)

Jiang, Y.-F., Cantiello, M., Bildsten, L., Quataert, E., & Blaes, O. 2015, ApJ, 813, 74, doi: [10.1088/0004-637X/813/1/74](https://doi.org/10.1088/0004-637X/813/1/74)

Jiang, Y.-F., Cantiello, M., Bildsten, L., et al. 2018, Nature, 561, 498, doi: [10.1038/s41586-018-0525-0](https://doi.org/10.1038/s41586-018-0525-0)

Langanke, K., & Martínez-Pinedo, G. 2000, Nuclear Physics A, 673, 481, doi: [10.1016/S0375-9474\(00\)00131-7](https://doi.org/10.1016/S0375-9474(00)00131-7)

Langer, N., & Kudritzki, R. P. 2014, A&A, 564, A52, doi: [10.1051/0004-6361/201423374](https://doi.org/10.1051/0004-6361/201423374)

- Lecoanet, D., & Quataert, E. 2013, MNRAS, 430, 2363, doi: [10.1093/mnras/stt055](https://doi.org/10.1093/mnras/stt055)
- Lecoanet, D., Cantiello, M., Quataert, E., et al. 2019, ApJL, 886, L15, doi: [10.3847/2041-8213/ab5446](https://doi.org/10.3847/2041-8213/ab5446)
- Oda, T., Hino, M., Muto, K., Takahara, M., & Sato, K. 1994, Atomic Data and Nuclear Data Tables, 56, 231, doi: [10.1006/adnd.1994.1007](https://doi.org/10.1006/adnd.1994.1007)
- Paxton, B., Bildsten, L., Dotter, A., et al. 2011, ApJS, 192, 3, doi: [10.1088/0067-0049/192/1/3](https://doi.org/10.1088/0067-0049/192/1/3)
- Paxton, B., Cantiello, M., Arras, P., et al. 2013, ApJS, 208, 4, doi: [10.1088/0067-0049/208/1/4](https://doi.org/10.1088/0067-0049/208/1/4)
- Paxton, B., Marchant, P., Schwab, J., et al. 2015, ApJS, 220, 15, doi: [10.1088/0067-0049/220/1/15](https://doi.org/10.1088/0067-0049/220/1/15)
- Paxton, B., Schwab, J., Bauer, E. B., et al. 2018, ApJS, 234, 34, doi: [10.3847/1538-4365/aaa5a8](https://doi.org/10.3847/1538-4365/aaa5a8)
- Paxton, B., Smolec, R., Schwab, J., et al. 2019, ApJS, 243, 10, doi: [10.3847/1538-4365/ab2241](https://doi.org/10.3847/1538-4365/ab2241)
- Pols, O. R., Tout, C. A., Eggleton, P. P., & Han, Z. 1995, MNRAS, 274, 964, doi: [10.1093/mnras/274.3.964](https://doi.org/10.1093/mnras/274.3.964)
- Potekhin, A. Y., & Chabrier, G. 2010, Contributions to Plasma Physics, 50, 82, doi: [10.1002/ctpp.201010017](https://doi.org/10.1002/ctpp.201010017)
- Rogers, F. J., & Nayfonov, A. 2002, ApJ, 576, 1064, doi: [10.1086/341894](https://doi.org/10.1086/341894)
- Rogers, T. M., Lin, D. N. C., McElwaine, J. N., & Lau, H. H. B. 2013, ApJ, 772, 21, doi: [10.1088/0004-637X/772/1/21](https://doi.org/10.1088/0004-637X/772/1/21)
- Saumon, D., Chabrier, G., & van Horn, H. M. 1995, ApJS, 99, 713, doi: [10.1086/192204](https://doi.org/10.1086/192204)
- Shiode, J. H., Quataert, E., Cantiello, M., & Bildsten, L. 2013, MNRAS, 430, 1736, doi: [10.1093/mnras/sts719](https://doi.org/10.1093/mnras/sts719)
- Simón-Díaz, S., Godart, M., Castro, N., et al. 2017, A&A, 597, A22, doi: [10.1051/0004-6361/201628541](https://doi.org/10.1051/0004-6361/201628541)
- Timmes, F. X., & Swesty, F. D. 2000, ApJS, 126, 501, doi: [10.1086/313304](https://doi.org/10.1086/313304)
- Yusof, N., Hirschi, R., Meynet, G., et al. 2013, MNRAS, 433, 1114, doi: [10.1093/mnras/stt794](https://doi.org/10.1093/mnras/stt794)